

Chenyuan Zhang

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EXPERIENCE

MuLIP Lab, Tufts University

Medford, MA

Research Assistant in Reinforcement Learning & Robotics

Jan 2026 - Present

Advisor: Prof. [Jivko Sinapov](#)

- Developed and deployed Inverse Reinforcement Learning (IRL) models to infer human reward functions from behavioral datasets.
- Achieved a success rate over 90% in robotic fetch-and-place tasks by implementing a baseline robot learning algorithm combining DDPG, Hindsight Experience Replay, and Prioritized Experience Replay.

Zhejiang Eagle Semiconductor Co., Ltd.

Zhuji, Zhejiang, China

Information Engineering Intern

Jan 2025 - Jul 2025

Reduced manual data processing time 50% by building a Python-based tool for chip-manufacturing workflows and enabling more timely monitoring of production status.

SELECTED PROJECTS

HAL — Assistive Task Agent [<https://hal-bay.vercel.app>]

Mar 2026 - May 2026

- Independently designed, built, and deployed a full-stack AI task-assistant (FastAPI + React) with a 3-stage LLM pipeline for intent classification, task extraction, and probability-aware planning advice.
- Designed a custom deadline-probability algorithm modeling multi-task interference via a closed-form integral; converges in <20 iterations, updating scores every second during active sessions.
- Built a time-aware recommendation module (Llama 3.3 70B via Groq) that proactively triggers advice when task finish probability drops across thresholds, with a live working timer feeding directly into real-time updates.

ARIA — Autonomous Revenue Integrity Agents [[live demo](#)] — YHack Project

Mar 2026

- Identified potential revenue leakage by building a multi-agent AI system by co-building a multi-agent AI system that detects contract-to-invoice revenue leakage across SaaS contracts, usage records, and invoices.
- Contributed to algorithm design and UI/UX for ranking recovery actions by dollar impact, urgency, and confidence score.

Intelligent Passenger Flow Analysis and Prediction for Urban Rail Transit [[link](#)]

Jan 2022 - May 2022

- Analyzed public outsourced-competition datasets containing anonymized metro trip records, station information, weather, holidays, and user attributes using MATLAB, SPSS.
- Built a time-series model based on a neural network to forecast future passenger flow across 8 rail lines and 168 stations with potential guidance value for urban rail operators.

Chenyuan Zhang, Xinyu Huang, Zheyu Fang. **Fungal and Ecosystem** [[link](#)]

Jan 2021

Mathematical Modeling Research Paper (M Award in MCM/ICM)

- Developed differential-equation models to quantify fungal decomposition rates ($R = 0.848$, $p < 0.01$).
- Predicted the relative strengths and weaknesses of each species and possible combinations of species based on Ecological Niche Modeling.

EDUCATION

New York University, Incoming M.S. in Computer Science

Sep 2026 - Dec 2027 (expected)

New York, NY

Tufts University, Post-Baccalaureate Certificate in Computer Science

Sep 2025 - Aug 2026

Medford, MA

Zhejiang Chinese Medical University B.Med., Clinical Medicine

Sep 2018 - Jun 2024

Hangzhou, China

AWARDS

Meritorious Winner (top 7% of all teams), Mathematical Contest in Modeling (MCM/ICM)

2021

National Undergraduate Mathematical Contest in Modeling (CUMCM) (Second Prize)

2020

Academic Competition Award, Zhejiang Chinese Medical University

2020

SKILLS

Languages: Chinese (Native Proficiency), English (Professional/Fluent)

Framework & Tools: React, PyTorch, TensorFlow, NumPy, OpenCV, Node.js, Git, Arduino

Programming Languages: C++, Python, JavaScript, MATLAB, SQL